IBRAHIM SAID HUSSEIN ELSAYED

Robotics & Embedded Systems Engineer

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PROFILE

Robotics and embedded systems engineer (Bachelor of Mechatronics Engineering with Honours, University of Wollongong Malaysia, Washington Accord) who works from microcontroller firmware and sensor PCBs up to ROS 2 and C++ on the robot. On live service-robot deployments I traced ROS communication bottlenecks causing 30%+ sensor-to-actuator lag; at Heralds I containerised an autonomy stack to run identically on 3+ edge servers and built DDS/MQTT telemetry with sub-100 ms latency.

WORK EXPERIENCE

Software Engineer – Systems Integration | Heralds Investments FZE LLC | Dubai, UAE | Nov 2025 – Sep 2026

- Designed modular hardware-software integration layers in C++ and Python, with standard interface contracts so teams could plug in new modules without rework.
- Containerised the full autonomy stack with Docker, giving one consistent deployment across 3+ edge server environments and removing environment-specific failures.
- Built real-time telemetry pipelines with DDS and MQTT, delivering system-health data across nodes in under 100 ms.

Robotics Engineer Intern – Field Operations | Robopreneur Sdn Bhd | Selangor, Malaysia | Jul 2025 – Sep 2025

- Validated service robots on site in live customer deployments, resolving ROS communication bottlenecks that caused 30%+ data lag in sensor-actuator pipelines.
- Diagnosed and patched C++ bugs in the navigation stack using field data.
- Wrote the system architecture and communication-interface documentation used to onboard new engineers.

Cybersecurity Intern – Network & Systems Security | SHRDC (Smart Factory) | Selangor, Malaysia | Jun 2024 – Aug 2024

- Installed Linux on 12 Raspberry Pi devices and configured a firewall on each to control inbound and outbound traffic.
- Designed subnetting to segment the devices, set up remote monitoring and control, and deployed Wazuh (SIEM) and Checkmk.

ENGINEERING PROJECTS

Cloud-Based IoT Indoor Air Quality Monitor – Final Year Project | Raspberry Pi 5, I²C, PCB design, Python, MariaDB, Node-RED | 2025

- Built a multi-point sensor network measuring CO_2, temperature, humidity (SCD41) and airflow (FS3000) across a university study room.
- Designed and fabricated 3 custom PCBs: dual TCA9548A I²C multiplexers to address identical sensors, P82B715 bus extenders for long cable runs, and regulated sensor nodes.
- Logged data to MariaDB through Python and built a live Node-RED dashboard; designed and wired the electrical enclosure.
- Ran a 3-day study benchmarked against WHO, ASHRAE 55, OSHA and ICOP IAQ 2010 (peak 31.0 °C vs a 23–26 °C comfort band; CO_2 up to 903 ppm).

Humanoid Robot – ROBOTIS Bioloid | ARM Cortex-M3 (CM-530) – assembled, wired and programmed a Bioloid humanoid robot.

Trash Collection & Classification Machine | Mechatronics, motor control, mechanical design | 2023

- Built a motor-driven, multi-tier rotating sieve that separates mixed waste by size, with a guide wall that sweeps sorted items into a collection bin.

Autonomous Navigation Stack (ROS 2) | C++, EKF sensor fusion, IMU + LiDAR

- Developed modular C++ navigation libraries with EKF-based IMU + LiDAR fusion for stable localisation in GPS-denied environments, with custom ROS 2 nodes for deterministic message delivery.

TECHNICAL SKILLS

- **Programming:** C++ (OOP), Embedded C, Python, Bash, CMake
- **Embedded & Hardware:** ESP32, ESP8266, Arduino, LPC2148, Raspberry Pi, ARM Cortex-M (CM-530), NVIDIA Jetson, Pixhawk (PX4), I²C, PCB design, IMU, LiDAR
- **Robotics & Autonomy:** ROS 2, ROS 1, SLAM, localisation, EKF sensor fusion, autonomous navigation, Gazebo, RViz, MoveIt
- **Middleware & Data:** DDS, MQTT, MAVLink, Node-RED, MariaDB / SQL
- **Tools:** Linux (Ubuntu), Git, Docker, CI/CD, SolidWorks, MATLAB
- **Networking & Security:** Subnetting, firewall configuration, Wazuh (SIEM), Checkmk

EDUCATION

Bachelor of Mechatronics Engineering with Honours | University of Wollongong Malaysia | 2019 – 2025

Accredited by the Engineering Accreditation Council (Board of Engineers Malaysia); recognised under the Washington Accord.

Secondary School Certificate – Scientific Track | A'shifaa National Secondary School, Riyadh | 2016 – 2018 | 99.88% (Excellent)

MEMBERSHIPS, CERTIFICATIONS & LANGUAGES

- Student Member, The Institution of Engineers, Malaysia (IEM) – 2020 – 2025
- IELTS Academic – Overall Band 7.5 (CEFR C1)
- Google Cloud: Introduction to Generative AI, Applying AI Principles · Anthropic: Claude 101, Claude Code 101
- Languages: Arabic (native) · English (C1) · Malay (professional working) | Interests: certified in bouldering and top-rope climbing